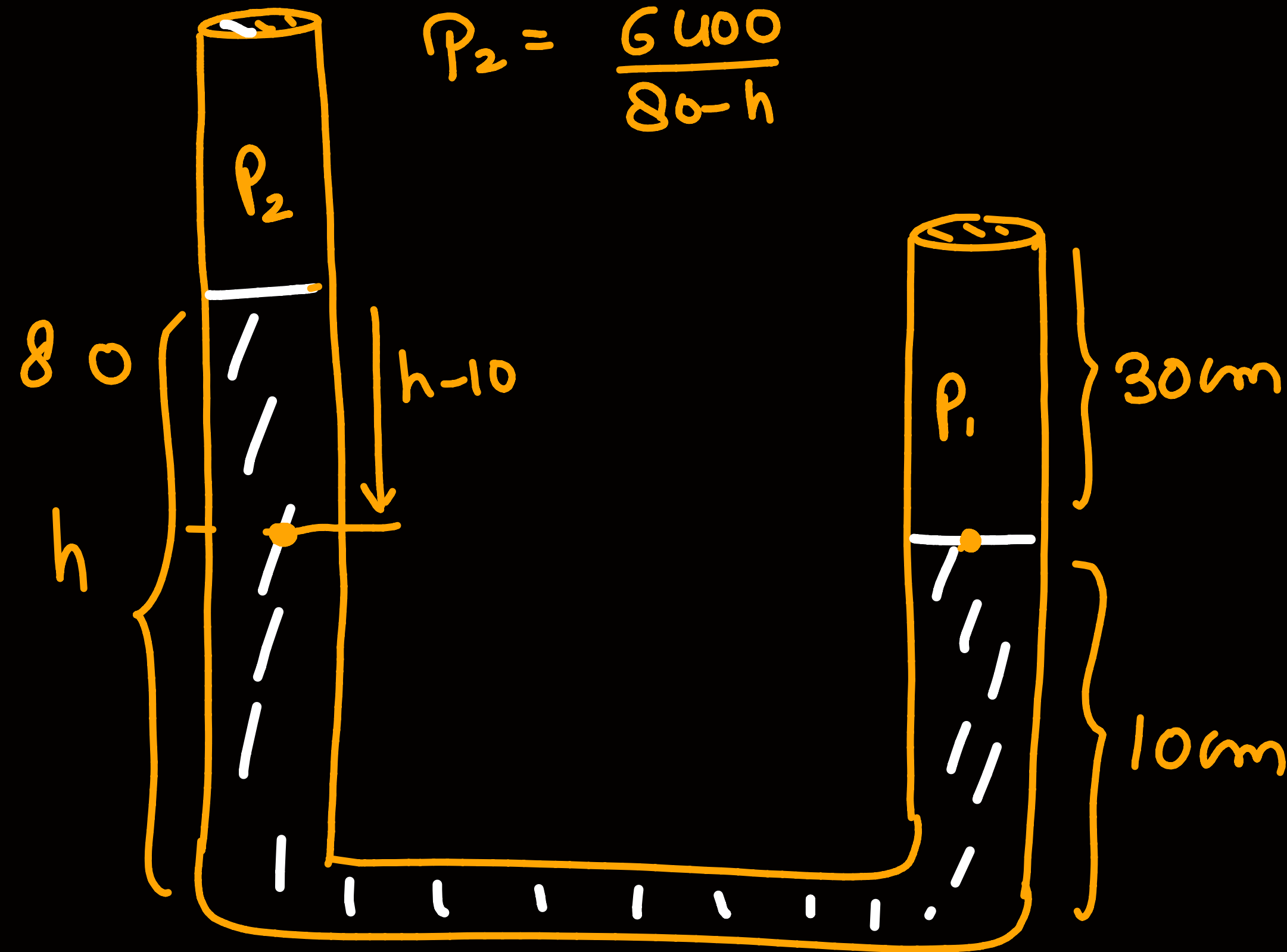
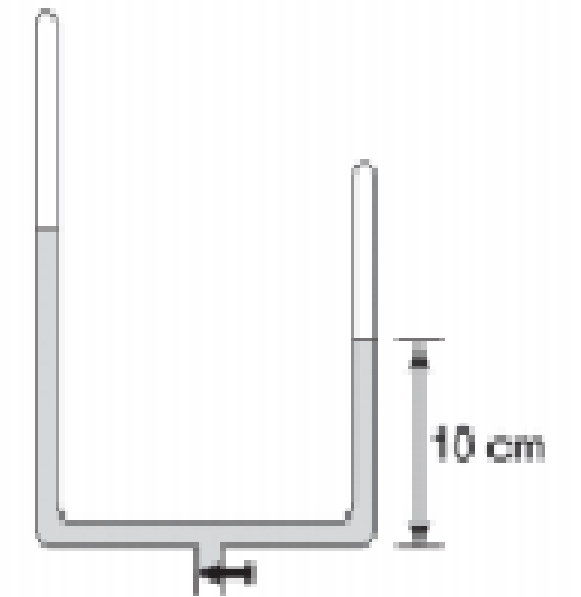


$$80 \times A \times 80 = P_2 A (80 - h) \quad P_0 = 80 \text{ cm of Hg}$$

$$P_2 = \frac{6400}{80 - h}$$



Q. 7: A U shaped tube has two arms of equal cross section and lengths $\ell_1 = 80$ cm and $\ell_2 = 40$ cm. The open ends are sealed with air in the tube at a pressure of 80 cm of mercury. Some mercury is now introduced in the tube through a stopcock connected at the bottom (the air is not allowed to leak out). In steady condition the length of mercury column in the shorter arm was found to be 10 cm. Find the length of the mercury column in the longer arm. Neglect the volume of the part of the tube connecting two arms and assume that the temperature is constant.



$$80 \times A \times 40 = P_1 \times A \times 30$$

$$P_1 = \frac{320}{3}$$

$$P_1 = \frac{320}{3} = P_2 + h - 10 = \frac{6400}{80 - h} + h - 10 = \frac{320}{3}$$

$$h = 16 \dots \text{cm}$$

$$M_{O_2} = 32 \text{ g/mole}$$

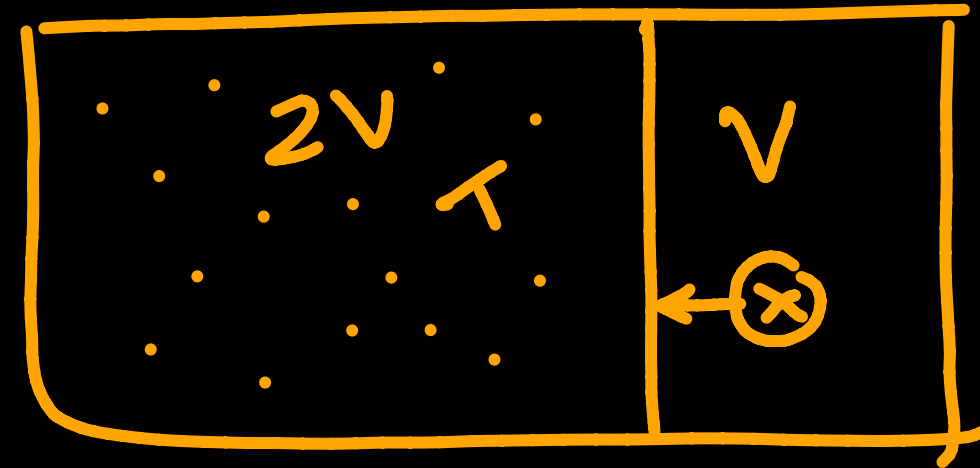
$$M_{N_2} = 28 \text{ g/mole}$$

$$U_0 = \sqrt{\frac{8RT_0}{\pi \cdot 32}}$$

$$\sqrt{\frac{8}{7}} U_0 = \sqrt{\frac{8RT_N}{\pi \cdot 28}}$$

$$\sqrt{\frac{8}{7}} = \sqrt{\frac{8 \cancel{32}}{7 \cdot 28} \frac{T_N}{T_0}}$$

$$\boxed{T_N = T_0}$$

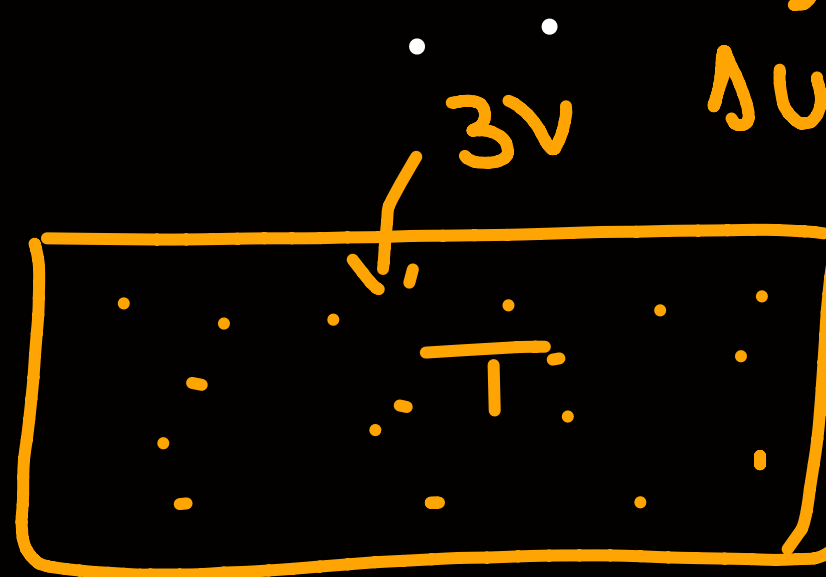


free expansion

$$W=0$$

$$\Delta Q=0$$

$$\Delta U=0$$



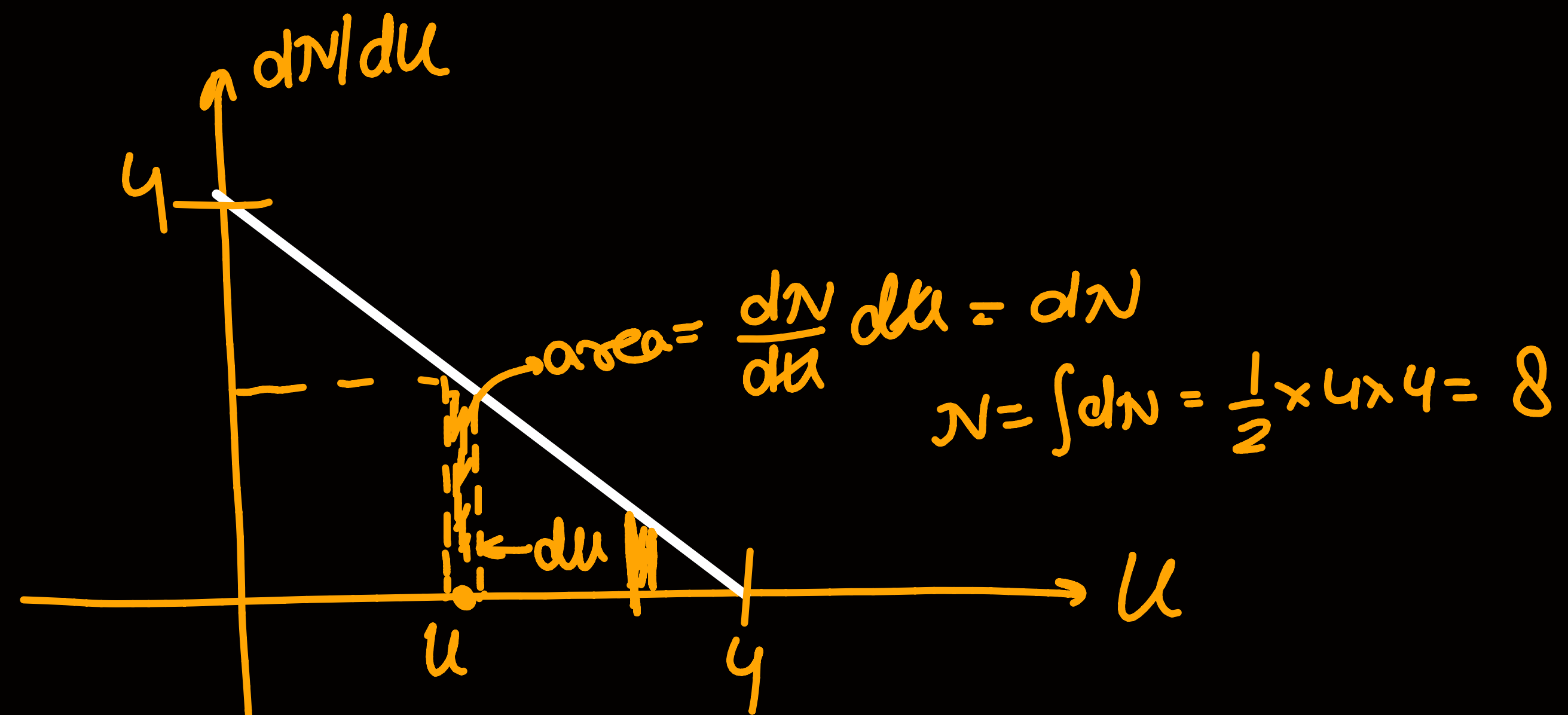
$$U = \sqrt{\frac{8RT}{\pi m}} = U_0$$

$$T_{mix} = \frac{n_1 C_{v1} T_1 + n_2 C_{v2} T_2}{n_1 C_{v1} + n_2 C_{v2}}$$

Q. 23: An insulated cylindrical vessel is divided into three identical parts by two partitions 1 and 2. The left part contains O_2 gas, the middle part has N_2 and the third chamber has vacuum. The average molecular speed in oxygen chamber is U_0 and that in nitrogen chamber is $\sqrt{\frac{8}{7}} U_0$. Pressure of the gases in two chambers is same. Partition 1 is removed and the gases are allowed to mix. Now the stopper holding the partition 2 is removed and it slides to the right wall of the container, so that the mixture of gases occupy the entire volume of the container.

Find the average speed of O_2 molecules now.

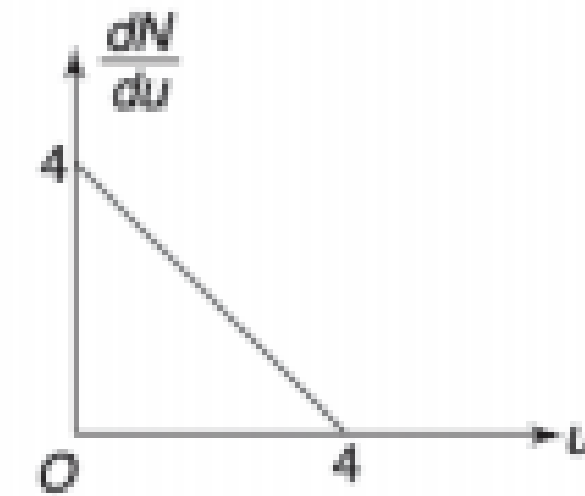
O_2	N_2	
1	2	



$$u_{rms} = \sqrt{\frac{\int u^2 dN}{\int dN}} = \sqrt{\frac{\int_0^4 u^2 (-u + 4) du}{8}}$$

$$= \sqrt{\frac{\int_0^4 (-u^3 + 4u^2) du}{8}} = \sqrt{\frac{\left[\frac{-u^4}{4} + \frac{4u^3}{3}\right]_0^4}{8}} = \sqrt{\frac{-64 + \frac{256}{3}}{8}} = \sqrt{\frac{64}{3 \times 8}} = \sqrt{\frac{8}{3}}$$

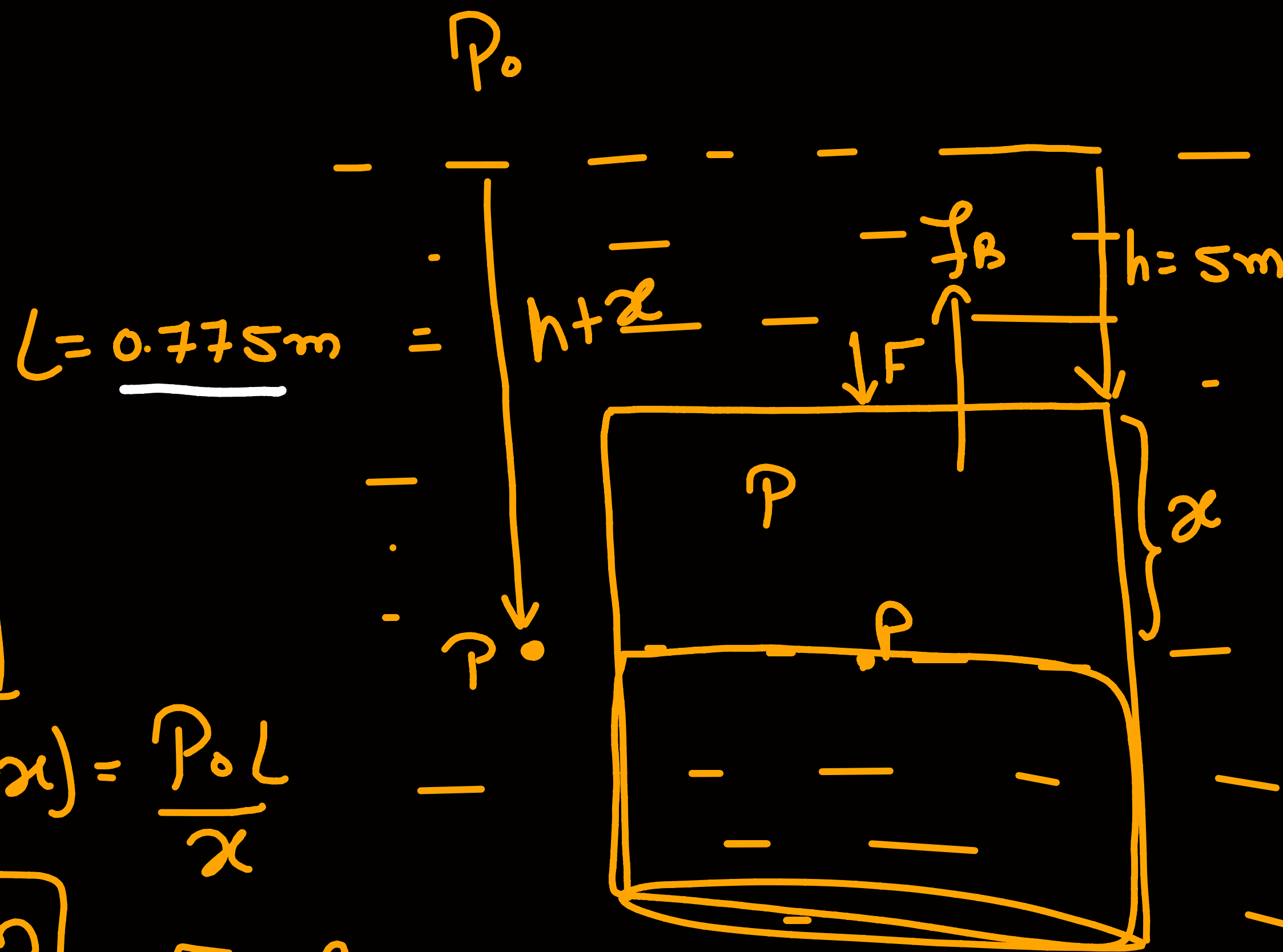
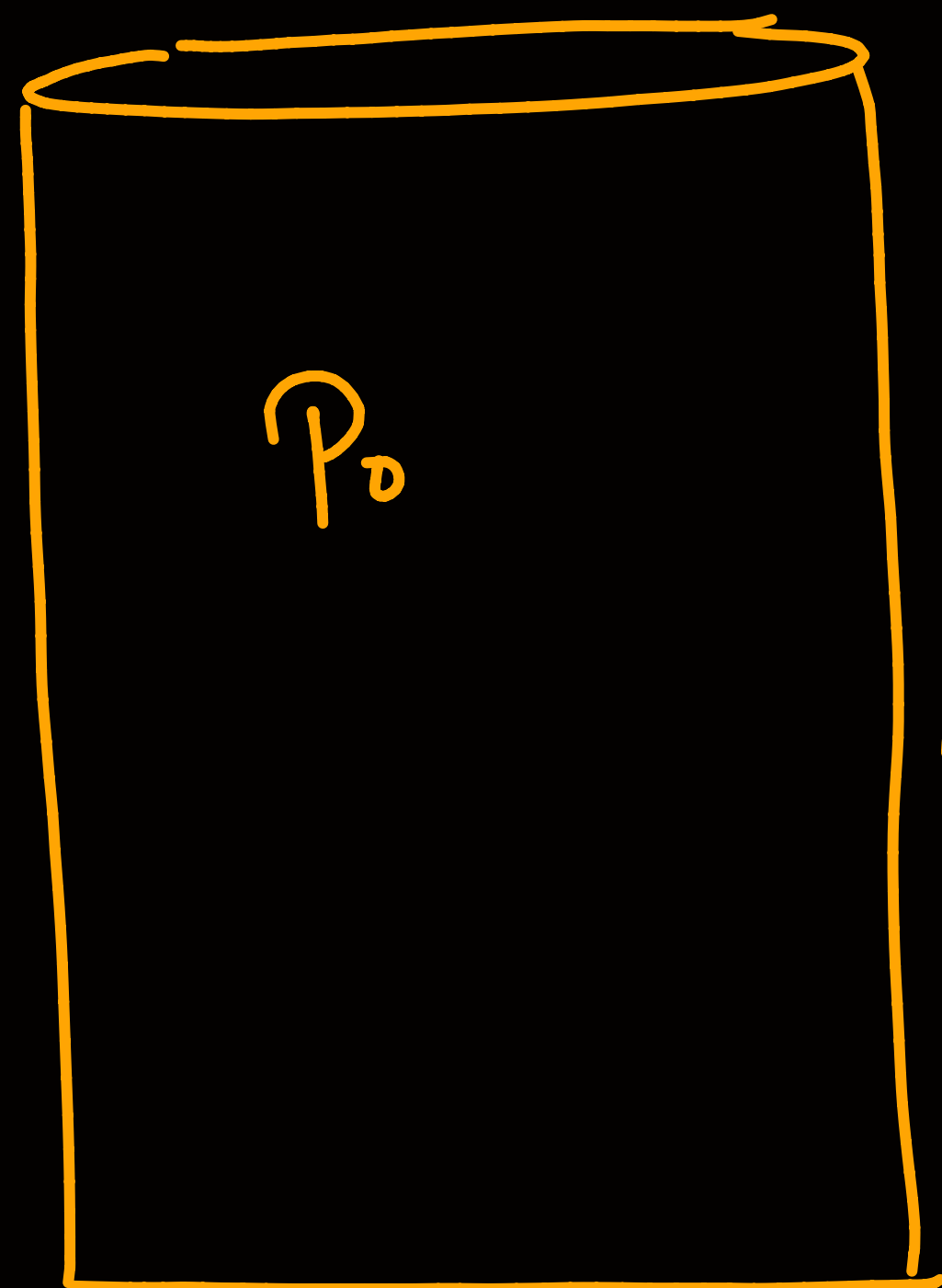
Q. 27: A hypothetical gas sample has its molecular speed distribution graph as shown in the figure. The speed (u) and $\frac{dN}{du}$ have appropriate units. Find the root mean square speed of the molecules. Do not worry about units.



$$y = mx + C \quad \begin{matrix} 256 \\ 192 \\ 64 \end{matrix}$$

$$\frac{dN}{du} = -u + 4$$

$$P_0 A L = P A x$$



$$P = P_0 + \rho g (h + x) = \frac{P_0 L}{x}$$

$$x = 0.5 \text{ m}$$

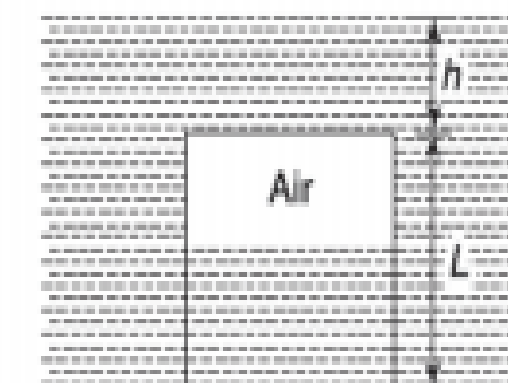
$$F = f_B$$

$$= 10^3 \times 0.05 \times 0.5 \times 10 = 250 \text{ Newton}$$

Q. 38: A cylindrical container has cross sectional area of $A = 0.05 \text{ m}^2$ and length $L = 0.775 \text{ m}$. Thickness of the wall of the container as well as mass of the container is negligible. The container is pushed into a water tank with its open end down. It is held in a position where its closed end is $h = 5.0 \text{ m}$ below the water surface. What force is required to hold the container in this position? Assume temperature of air to remain constant.

Atmospheric pressure $P_0 = 1 \times 10^5 \text{ Pa}$; Acceleration due to gravity $g = 10 \text{ m/s}^2$

Density of water $= 10^3 \text{ kg/m}^3$



$$\underline{1 \rightarrow 2} \quad \underline{P \propto T} \quad V = \text{const} \quad W = 0 \quad \text{③-9}$$

$$2 \rightarrow 3 \quad P = \text{const} \quad W = P \Delta V = nRT \\ = nR(T_3 - T_2)$$

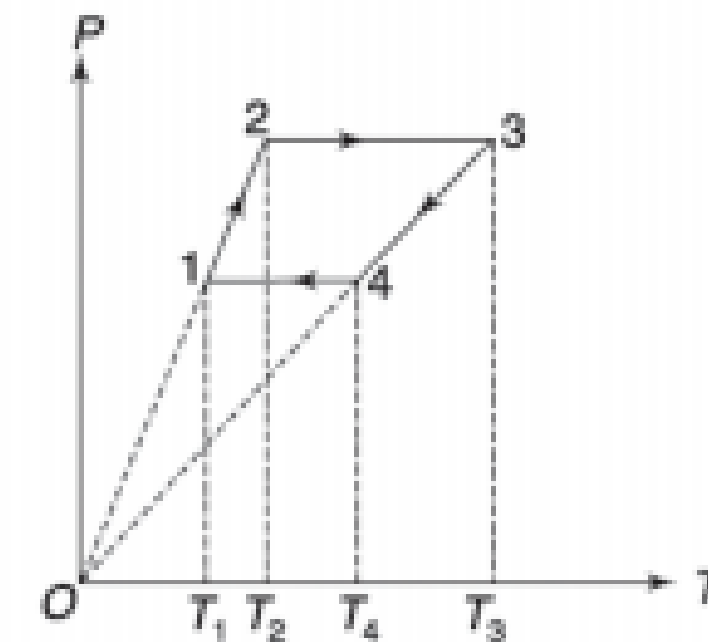
$$4 \rightarrow 1 \quad W = nRT \\ = nR(T_1 - T_4)$$

$$W = nR(T_1 + T_3 - T_2 - T_4)$$

O₂ gas γ.1.

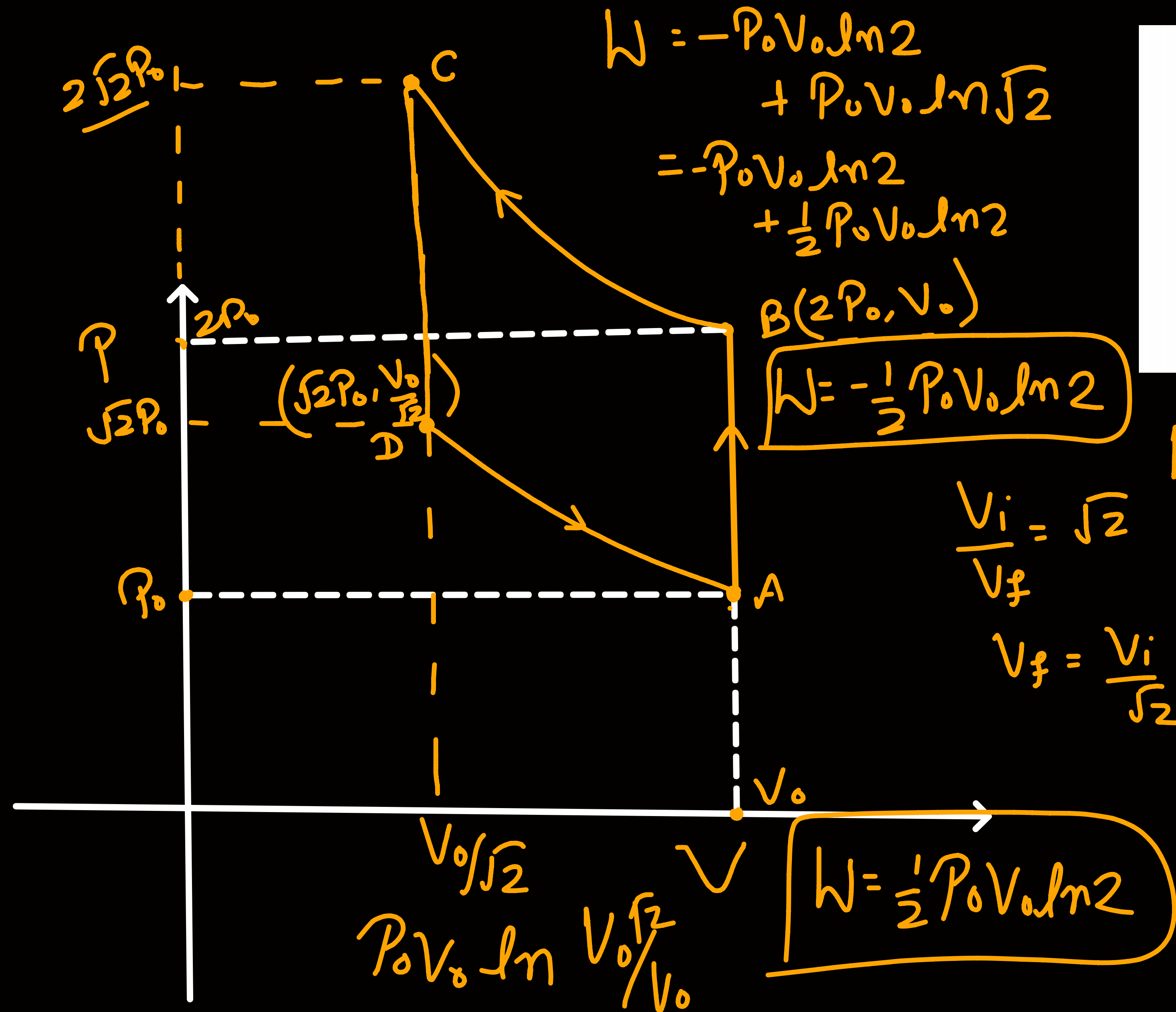
First Law of thermodynamics

Q. 11: n moles of an ideal gas is taken through a four step cyclic process as shown in the diagram. Calculate work done by the gas in a cycle in terms of temperatures T_1 , T_2 , T_3 and T_4 .

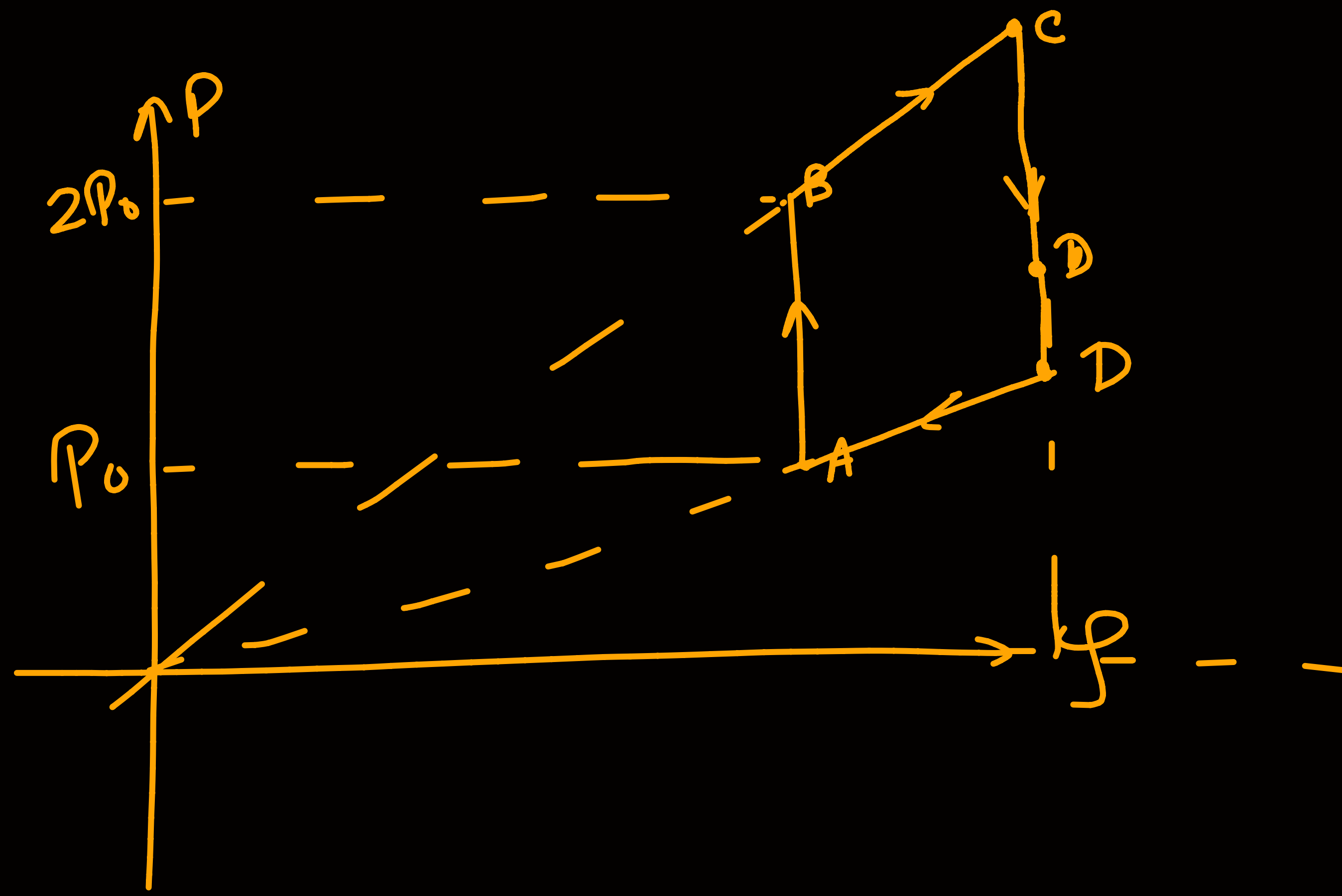


Q. 41: An ideal gas is taken through a thermodynamic cycle ABCDA. In state A pressure and volume are P_0 and V_0 respectively. During the process $A \rightarrow B$, work done by the gas is zero and its temperature increases two fold. During the process $B \rightarrow C$, internal energy of the gas remains constant but work done by it is $W_{BC} = -P_0 V_0 \ln 2$. In the process $C \rightarrow D$, the temperature decreases by 50% while the volume does not change. In the process $D \rightarrow A$ the temperature of the gas does not change.

- (a) Draw pressure - volume ($P-V$) and pressure - density ($P-\rho$) graph for the cyclic process.
(b) Calculate work done on the gas during the cycle.



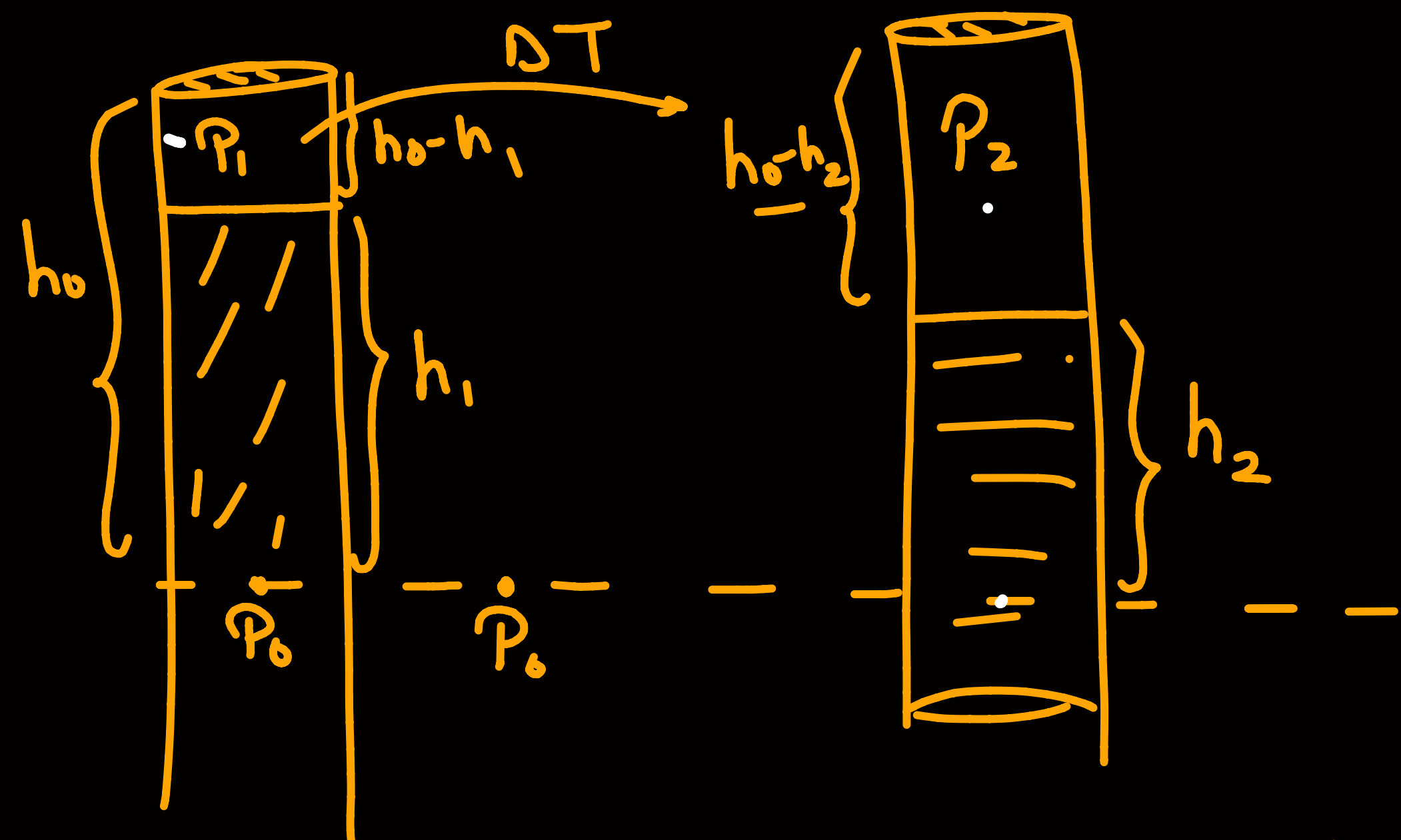
$$\begin{aligned}
 nRT &= PV \\
 W_{BC} &= -P_0 V_0 \ln 2 \\
 W &= PV \ln \frac{V_f}{V_i} \\
 &= 2P_0 V_0 \ln \frac{V_f}{V_i} \\
 &= P_0 V_0 \ln \left(\frac{V_f}{V_i} \right)^2 \\
 &= -P_0 V_0 \ln \left(\frac{V_i}{V_f} \right)^2
 \end{aligned}$$



$$P_M = f_{RT}$$

$$\Delta Q = \Delta U + W$$

$$= \frac{3}{2} \times 1 \times R \Delta T + W = \frac{3}{2} R \Delta T + W$$

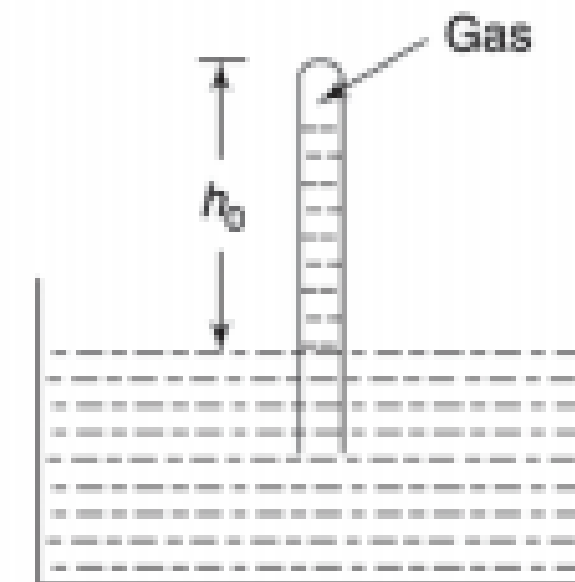


$$P_1 = P_0 - \rho g h_1$$

$$= \rho g (h_0 - h_1)$$

$$P_2 = \rho g (h_0 - h_2)$$

Q. 44: A glass tube is inverted and dipped in mercury as shown. One mole of an ideal monoatomic gas is trapped in the tube and the tube is held so that length of the tube above the mercury level is always h_0 meter. The atmospheric pressure is equal to h_0 meter of mercury. The mercury vapour pressure, heat capacity of mercury, tube and the container are negligible. How much heat must be supplied to the gas inside the tube so as to increase its temperature by ΔT ?

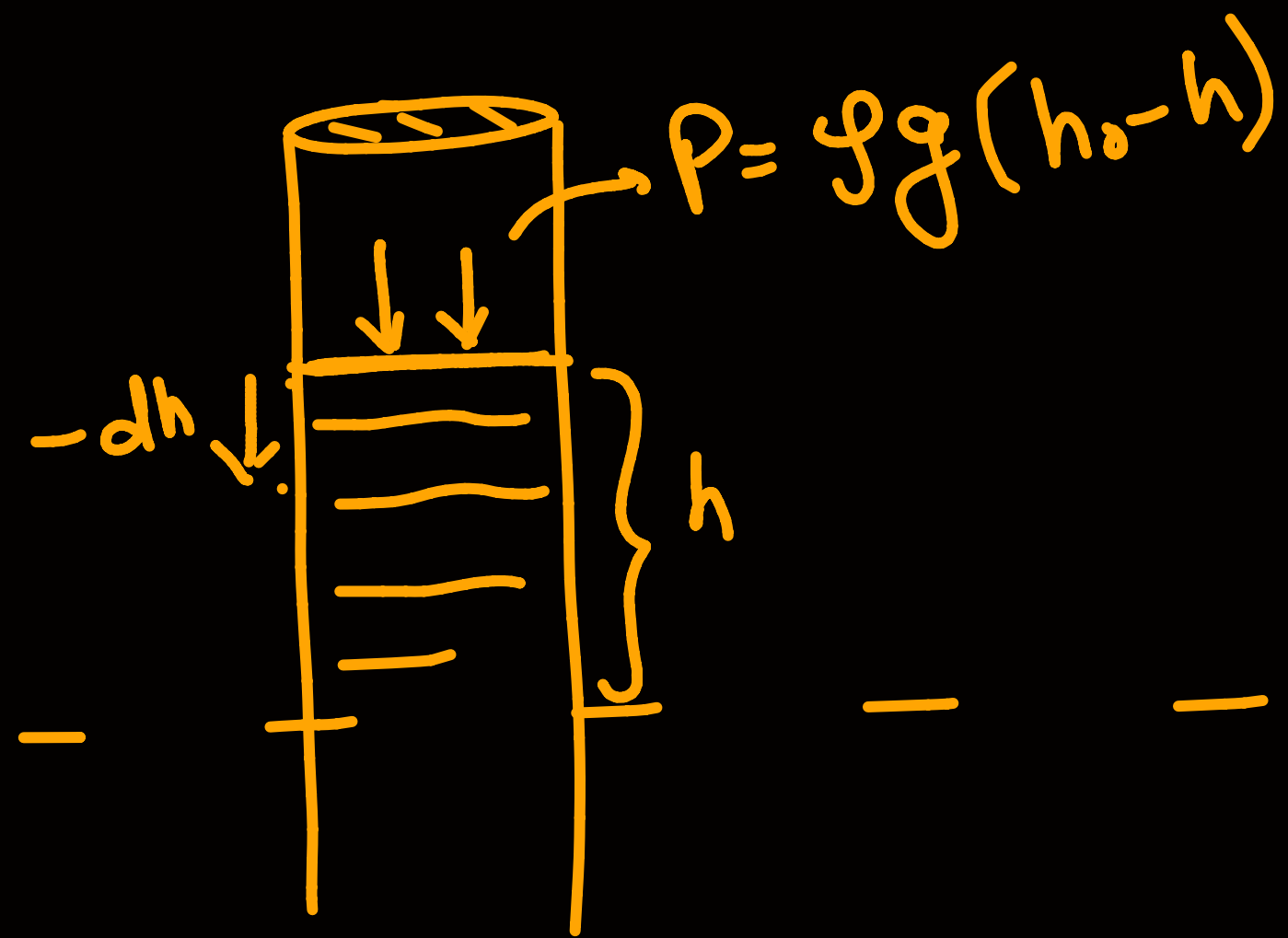


$$P_f V_f - P_i V_i = n R \Delta T$$

$$\rho g (h_0 - h_2)^2 A - \rho g (h_0 - h_1)^2 A = \Delta T$$

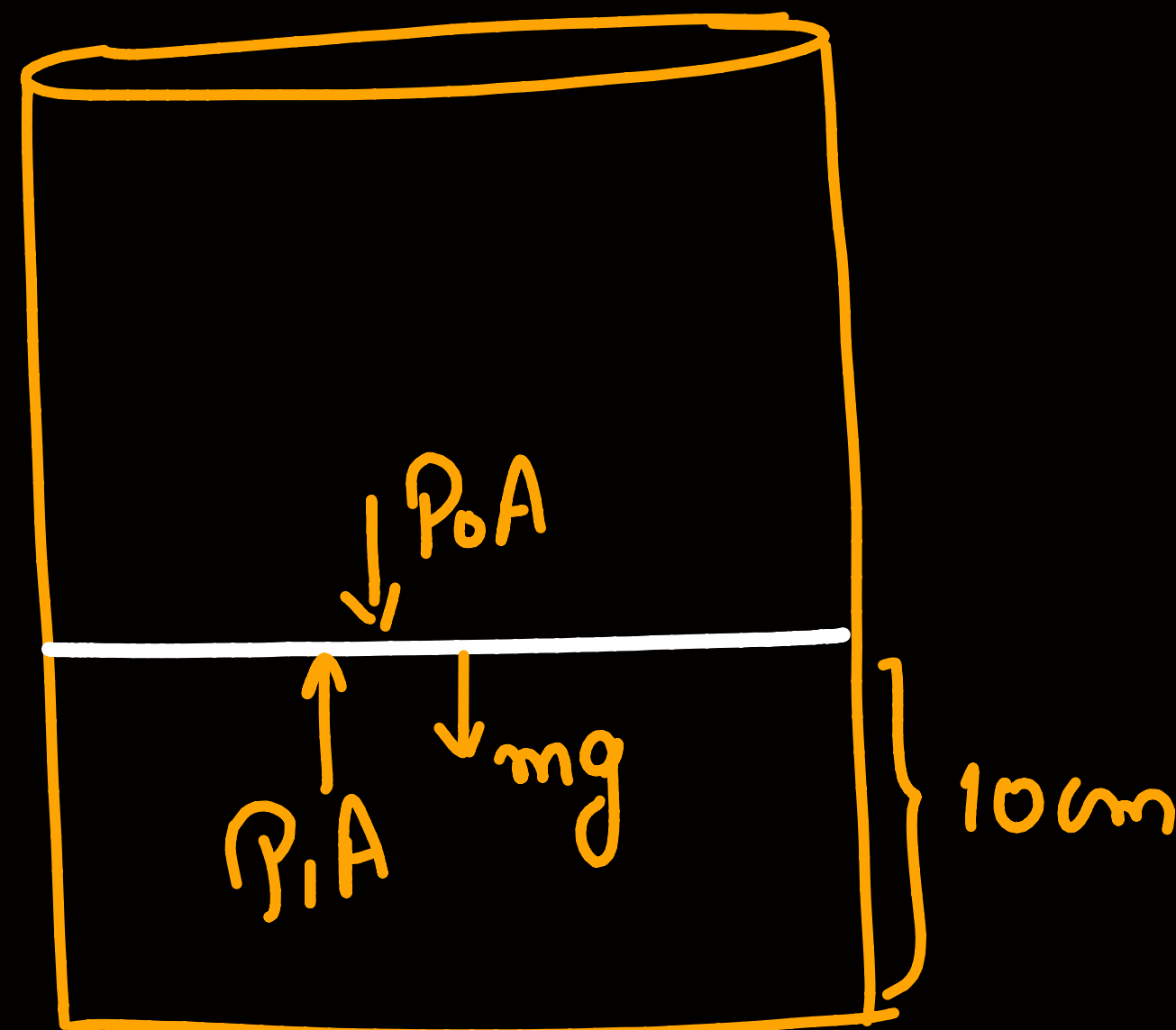
$$1 \times R$$

$$(h_0 - h_2)^2 - (h_0 - h_1)^2 = \frac{R \Delta T}{\rho g A}$$

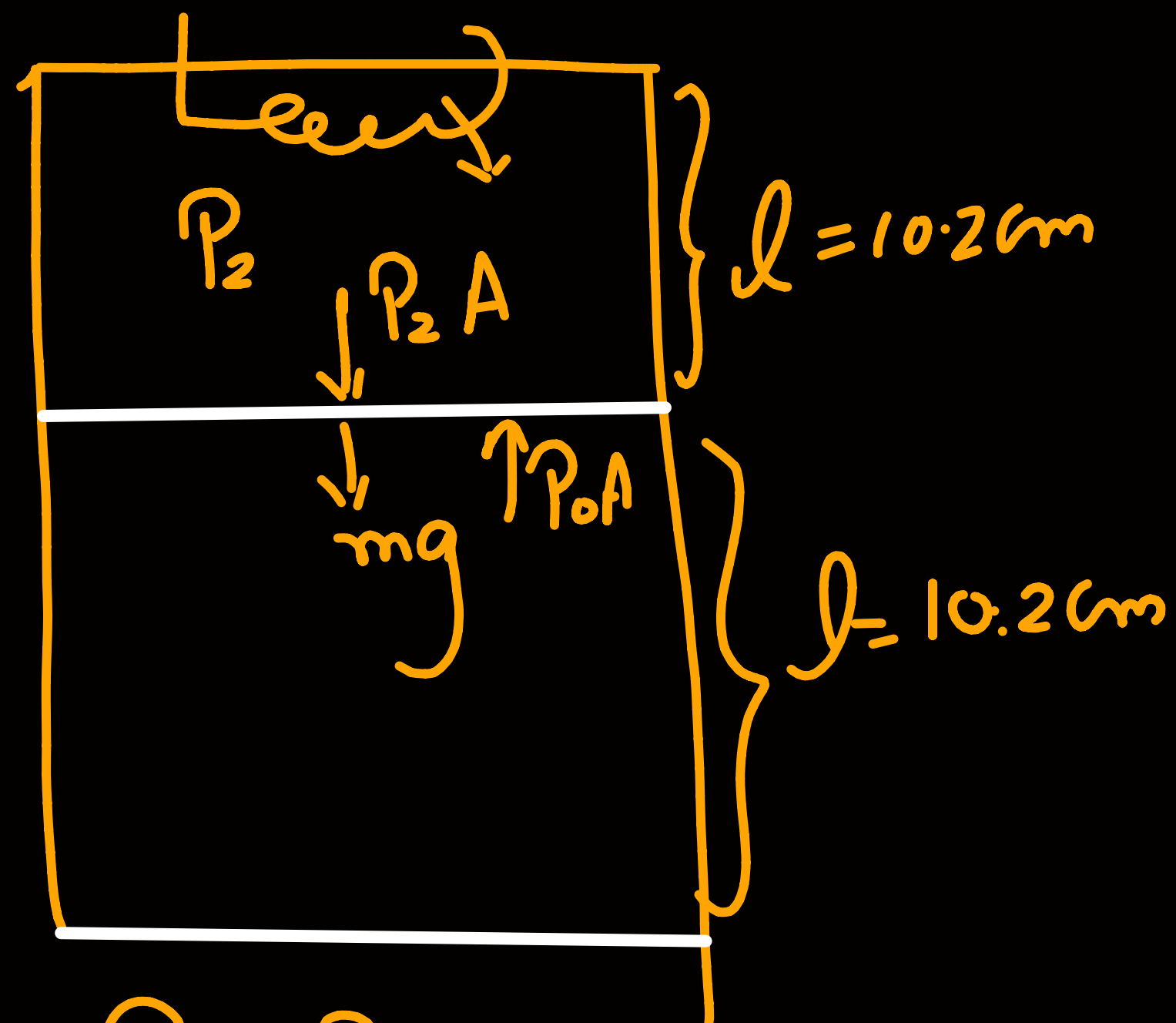


$$\begin{aligned}\Delta Q &= \Delta U + W \\ &= \frac{3}{2} R \Delta T + \frac{R \Delta T}{2} \\ &= \underline{2 R \Delta T}\end{aligned}$$

$$\begin{aligned}dw &= - \rho g (h_0 - h) A dh \\ W &= \int dw = - \rho g A \int_{h_1}^{h_2} (h_0 - h) dh \\ &= \rho g A \left[\frac{(h_0 - h)^2}{2} \right]_{h_1}^{h_2} \\ &= \frac{\rho g A}{2} \left\{ (h_0 - h_2)^2 - (h_0 - h_1)^2 \right\} \\ &= \frac{\cancel{\rho g A}}{2} \cdot \frac{R \Delta T}{\cancel{\rho g A}} = \frac{R \Delta T}{2}\end{aligned}$$



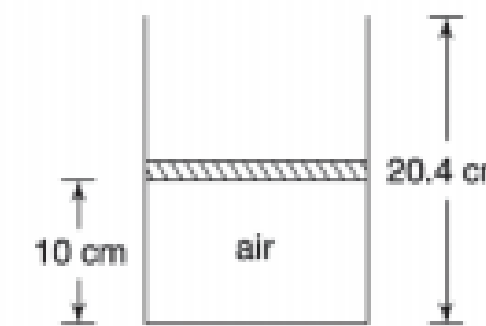
$$\begin{aligned}
 P_1 &= P_0 + \frac{mg}{A} \\
 &= 1.01 \times 10^5 + \frac{0.1 \times 10}{10 \times 10^{-4}} \\
 &= 1.01 \times 10^5 + 10^3 \\
 &= 1.02 \times 10^5 \text{ N/m}^2
 \end{aligned}$$



$$\begin{aligned}
 P_2 &= P_0 - \frac{mg}{A} \\
 &= 10^5 \text{ N/m}^2
 \end{aligned}$$

$$\begin{aligned}
 \Delta Q &= W + \Delta U \\
 &=
 \end{aligned}$$

Q. 47: In the arrangement shown in figure, the piston can smoothly move inside the cylinder. The mass of the piston is $m = 100 \text{ g}$ and its cross sectional area is $A = 10 \text{ cm}^2$. The length of air column at temperature of $T = 27^\circ\text{C}$ is 10 cm . Overall length of the cylinder is 20.4 cm . The container is turned upside down and the length of the air column in equilibrium was found to be l at 27°C .



Take $R = \frac{25}{3} \text{ J mol}^{-1} \text{ K}^{-1}$ and assume air to be diatomic gas.

$g = 10 \text{ m/s}^2$, atmospheric pressure is $1.01 \times 10^5 \text{ Nm}^{-2}$

(a) Find l

(b) If the air in the container is supplied heat in upside down position, the piston slowly begins to move down, and ultimately it gets ejected out of the cylinder. Calculate the amount of heat that the air must absorb for the piston to come out.

$$\begin{aligned}
 1.02 \times 10^5 \text{ N/m}^2 \times 10 \text{ cm} &= 10^5 \text{ N/m}^2 \times A \times l \\
 \boxed{l = 10.26 \text{ cm}}
 \end{aligned}$$

$$W + \Delta U = 10.2 \text{ J} + \Delta U = 10.2 \text{ J} + 25.5 \text{ J} = 35.7 \text{ J}$$

$$W = 10^5 \frac{\text{N}}{\text{m}^2} \times 10 \times 10^{-4} \times 10.2 \times 10^{-2}$$

$$= \underline{10.2 \text{ J}}$$

$$\Delta U = \frac{5}{2} n R \Delta T = \frac{5}{2} P \Delta V$$

$$= \frac{5}{2} \times 10^5 \times 10 \times 10^{-4} \times 10.2 \times 10^{-2}$$

$$= 5 \times 5.1 \text{ J} = 25.5 \text{ J}$$